

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **agentic AI systems** using Python, focusing on **conversational interfaces** and integrating with **APIs and microservices**, processing over 5,000 user interactions daily.
- Developed and optimized **AI pipelines** for real-time and asynchronous interactions, ensuring efficient data flow for **LLM-powered agents** and reducing processing latency by **18%**.
- Contributed to the development of **LLM-powered agents** capable of reasoning and task execution, incorporating **prompt engineering** techniques to enhance response accuracy for complex queries.
- Wrote clean, maintainable, and efficient **Python code** and comprehensive **unit tests**, achieving over **90% code coverage** for new feature development.
- Collaborated with **AI researchers and senior engineers** to integrate innovative AI solutions into production, participating in sprint planning and technical estimation for new feature development.
- Optimized **AI models and workflows** for performance and scalability within a **cloud environment**, reducing infrastructure costs by **12%** through resource tuning.
- Containerized **AI services** using **Docker** and deployed to a **Kubernetes cluster (EKS)**, managing configuration and resource allocation for predictable rollouts.

SKILLS

Programming Languages – Python, Java, SQL, R

AI/ML Frameworks – TensorFlow, PyTorch, Hugging Face, scikit-learn, LangChain, Bedrock

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch), GCP, Azure

Backend Development – Spring Boot, REST APIs, Microservices, FastAPI, Flask

Data Tools & Databases – PostgreSQL, MySQL, Oracle, pandas, NumPy, Power BI, Tableau

DevOps & Orchestration – Docker, Kubernetes, Jenkins, Git, Linux

PROJECTS

Generative AI Chatbot for Customer Support Automation

- Developed a **Python-based chatbot** leveraging **LLMs** (e.g., GPT models) and **LangChain** to automate customer support inquiries, handling an average of **2,000 queries per day**.
- Implemented **prompt engineering** strategies and RAG (Retrieval-Augmented Generation) to improve response relevance and accuracy for common customer issues, reducing human agent escalation by **30%**.
- Designed and deployed the chatbot as a microservice on **AWS Lambda** integrated with a **FastAPI** interface, ensuring scalability and low latency for real-time interactions.

AI-Powered Agentic System for Task Execution

- Engineered an **agentic AI system** in Python to autonomously execute predefined tasks based on user input, integrating with **APIs** for external service interaction.
- Built a scalable **AI pipeline** using **scikit-learn** for data pre-processing and feature extraction from user inputs, enabling the agent to process over **1,500 task requests** weekly.
- Containerized the agent system with **Docker** and managed deployments on a **Kubernetes** cluster, ensuring high availability and efficient resource utilization.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)